

APSC 131 – Chemistry of Materials

Lecture 02 – Introduction to Principles of Chemical Reactivity

Learning Plan

Keywords

- Periodic Table
- Electronegativity
- Valence

Resources

- See curated OnQ video list

Learning Outcomes

- Recall prior knowledge of the periodic table, with emphasis on common valencies and oxidation states.

Chemistry and Society's Grand Challenges

What high school chemistry topics can you recall?

What technical / social / economic problems are you interested in?

Natural Sciences versus Engineering Sciences

	Natural Science	Engineering Design Science
Objectives	To explain natural phenomena and create general theories.	To create objects, systems, and processes that solve practical problems.
Methodologies	Empirical observation, hypothesis testing, controlled experiments, statistical analysis.	Iterative design, prototyping, evaluation, and refinement; often includes user feedback and contextual analysis.
Outcomes	Theories, laws, and models that describe how the world works.	Innovative artifacts and frameworks that improve real-world practices.
Nature of Knowledge	Descriptive and explanatory; seeks truth and objectivity.	Prescriptive and constructive; seeks utility and effectiveness.
Evaluation Criteria	Validity, reliability, reproducibility, generalizability.	Utility, usability, effectiveness, and relevance to context.

Chemistry – The Central Science

Chemistry is hypothesis-driven, descriptive and reductionist.

- In natural and social science, the goal is to explore, describe, explain, and predict.
- Problems are narrowly defined to ensure that a hypothesis can be tested thoroughly and reliably.

CHEMISTRY (SCIENCE)



- Studies redox reactions in batteries



- Investigates ion transport and electrode materials
- Focuses on understanding electrochemical mechanisms

Engineering – Chemistry with Purpose

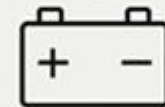
Engineering is design-driven, application focused and systemic

- The purpose of design science is to produce something that does not yet exist, or to modify existing solutions to achieve better results.
- Its methods are creative, iterative, and often experimental. This is ideal if you want to try things that haven't been done before.

ENGINEERING CHEMISTRY (DESIGN SCIENCE)

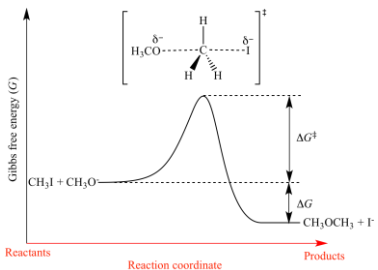
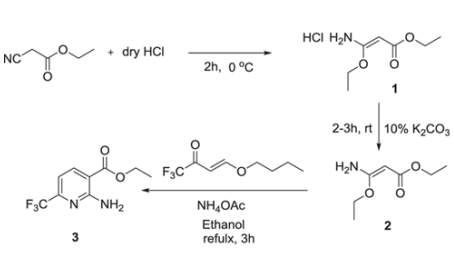
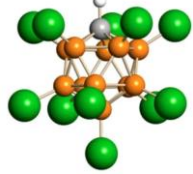
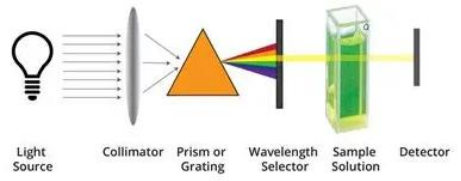
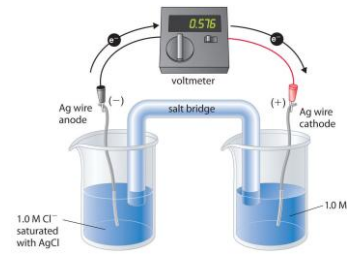


- Designs battery systems for scalability and safety



- Optimizes manufacturing processes for energy storage devices
- Integrates systems into real-world applications (e.g, electric vehicles)

Branches of Chemistry Covered in APSC 131/132

Physical	Organic	Inorganic	Analytical	Electrochemical
				
Thermodynamics Reaction Kinetics Quantum Mechanics Statistical Mechanics	Synthesis of organic (C,H,N,O) molecules Polymer chemistry	Ionic, organometallic, and solid-state compounds, minerals	Spectroscopy / spectrometry, Potentiometry, Chromatography	Galvanic cells, Electrolytic cells, Fuel cells
Chemical reaction engineering, material selection/design	Pharmaceutical and agrochemical synthesis, fine chemical production	Ceramics, glasses, semi-conductors, photo-voltaics, metallurgy, catalysis	Magnetic resonance imaging, chemical diagnostics, forensic and security applications	Energy storage, H ₂ production, electroplating, corrosion protection

+ biochemistry, medicinal chemistry, environmental chemistry, green chemistry + ...

APSC 131 Unit 1 – Principles of Chemical Reactivity

Lecture	Topics
2	Chemistry in an engineering context
3	Fundamental concepts
4	Atomic structure and models of covalent bonding
5	Chemical reactivity – thermodynamics
6	Chemical reactivity – kinetics
7	Acid-base chemistry
8	Ionic reactions in solution
9	Nucleophiles and electrophiles
10	Organic process development I - mechanisms
11	Organic process development II - mechanisms
12	Test your knowledge review

Periodic Table of the Elements

11.008																		24.003																	
H																		He																	
Hydrogen																		Helium																	
36.94		49.012																1020.180																	
Li		Be																Ne																	
Lithium		Beryllium																Neon																	
1122.990		1224.305																1839.948																	
Na		Mg																Ar																	
Sodium		Magnesium																Argon																	
1939.098		2040.078		2144.956		2247.867		2350.942		2451.996		2554.938		264.000		2758.933		2858.693		2963.546		3065.38		3169.723		3272.63		3374.922		3478.96		3579.904		3683.798	
K		Ca		Sc		Ti		V		Cr		Mn		Fe		Co		Ni		Cu		Zn		Ga		Ge		As		Se		Br		Kr	
Potassium		Calcium		Scandium		Titanium		Vanadium		Chromium		Manganese		Iron		Cobalt		Nickel		Copper		Zinc		Gallium		Germanium		Arsenic		Selenium		Bromine		Krypton	
3785.468		3887.62		3988.906		4091.224		4192.906		4295.96		43(98)		44101.07		45102.91		46106.42		47107.87		48112.41		49114.82		50118.71		51121.76		52127.60		53126.90		54131.29	
Rb		Sr		Y		Zr		Nb		Mo		Tc		Ru		Rh		Pd		Ag		Cd		In		Sn		Sb		Te		I		Xe	
Rubidium		Strontium		Yttrium		Zirconium		Niobium		Molybdenum		Technetium		Ruthenium		Rhodium		Palladium		Silver		Cadmium		Indium		Tin		Antimony		Tellurium		Iodine		Xenon	
55132.91		56137.33		57-71		72178.49		73180.95		74183.84		75186.21		76190.23		77192.22		78195.08		79196.97		80200.59		81204.38		82207.2		83208.98		84(209)		85(210)		86(222)	
Cs		Ba		Lanthanides		Hf		Ta		W		Re		Os		Ir		Pt		Au		Hg		Tl		Pb		Bi		Po		At		Rn	
Caesium		Barium				Hafnium		Tantalum		Tungsten		Rhenium		Osmium		Iridium		Platinum		Gold		Mercury		Thallium		Lead		Bismuth		Polonium		Astatine		Radon	
87(223)		88(226)		89-103		104(267)		105(268)		106(269)		107(270)		108(269)		109(278)		110(281)		111(281)		112(285)		113(286)		114(289)		115(288)		116(293)		117(294)		118(294)	
Fr		Ra		Actinides		Rf		Db		Sg		Bh		Hs		Mt		Ds		Rg		Cn		Uut		Fl		Uup		Lv		Uus		Uuo	
Francium		Radium				Rutherfordium		Dubnium		Seaborgium		Bohrium		Hassium		Meitnerium		Darmstadtium		Roentgenium		Copernicium		Ununtrium		Flerovium		Ununpentium		Livermorium		Ununseptium		Ununoctium	

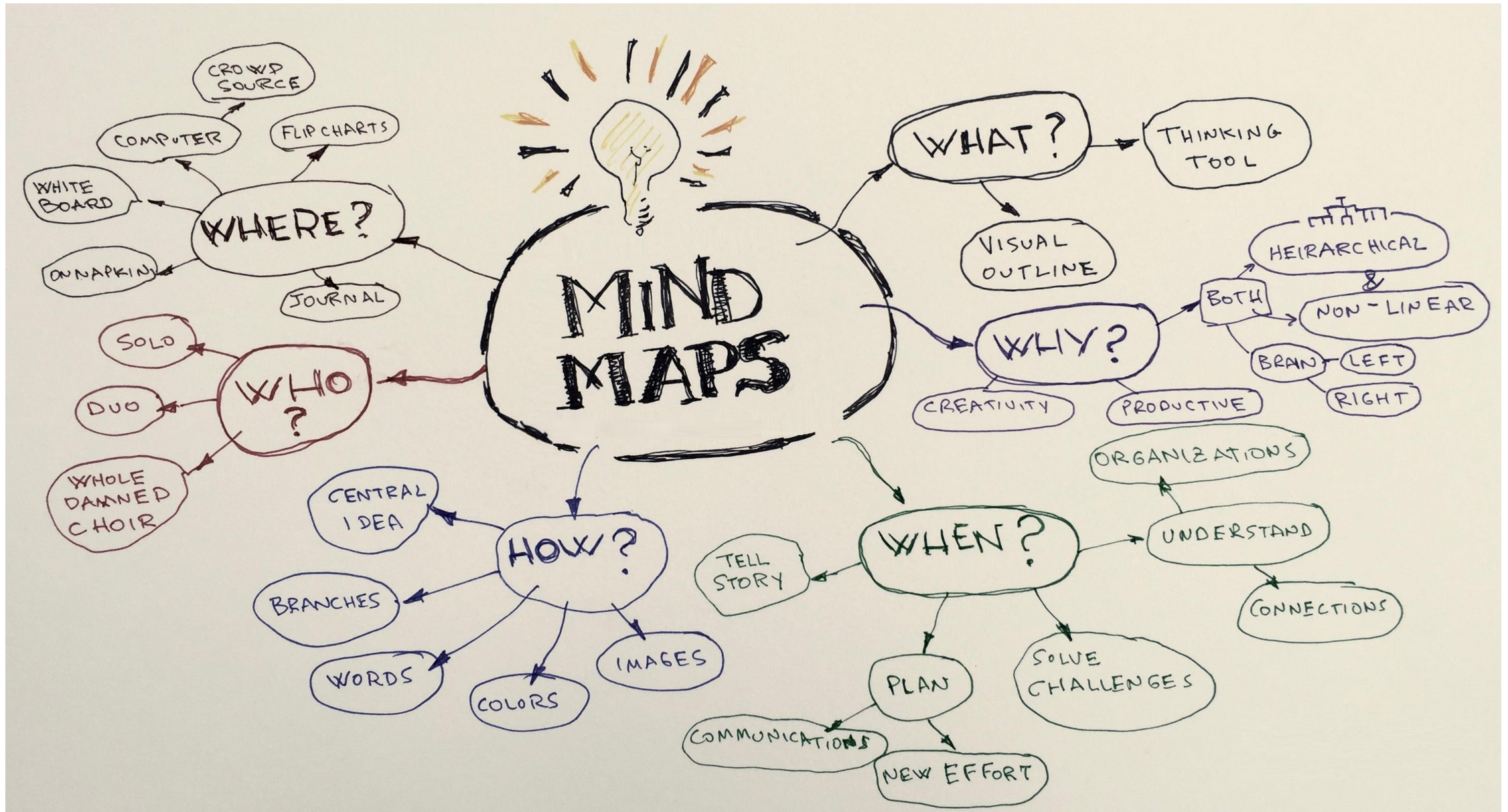
57 138.91 La Lanthanum	58 140.12 Ce Cerium	59 140.91 Pr Praseodymium	60 144.24 Nd Neodymium	61 (145) Pm Promethium	62 150.36 Sm Samarium	63 151.96 Eu Europium	64 157.25 Gd Gadolinium	65 158.93 Tb Terbium	66 162.50 Dy Dysprosium	67 164.93 Ho Holmium	68 167.26 Er Erbium	69 168.93 Tm Thulium	70 173.05 Yb Ytterbium	71 174.97 Lu Lutetium
89 (227) Ac Actinium	90 232.04 Th Thorium	91 231.04 Pa Protactinium	92 238.03 U Uranium	93 (237) Np Neptunium	94 (244) Pu Plutonium	95 (243) Am Americium	96 (247) Cm Curium	97 (247) Bk Berkelium	98 (251) Cf Californium	99 (252) Es Einsteinium	100 (257) Fm Fermium	101 (258) Md Mendelevium	102 (259) No Nobelium	103 (262) Lr Lawrencium

Metals	Alkali Metals
	Alkali Earth Metals
	Transition Metals
	Other Metals
	Lanthanides
	Actinides
Nonmetals	Metalloids
	Other Non Metals
	Halogens
	Noble Gases
	Unconfirmed

Atomic Number
Black = Solid
Red = Liquid
Blue = Gas
Grey = Unknown
At 0°C 1 bar

26 55.845
Fe
Iron

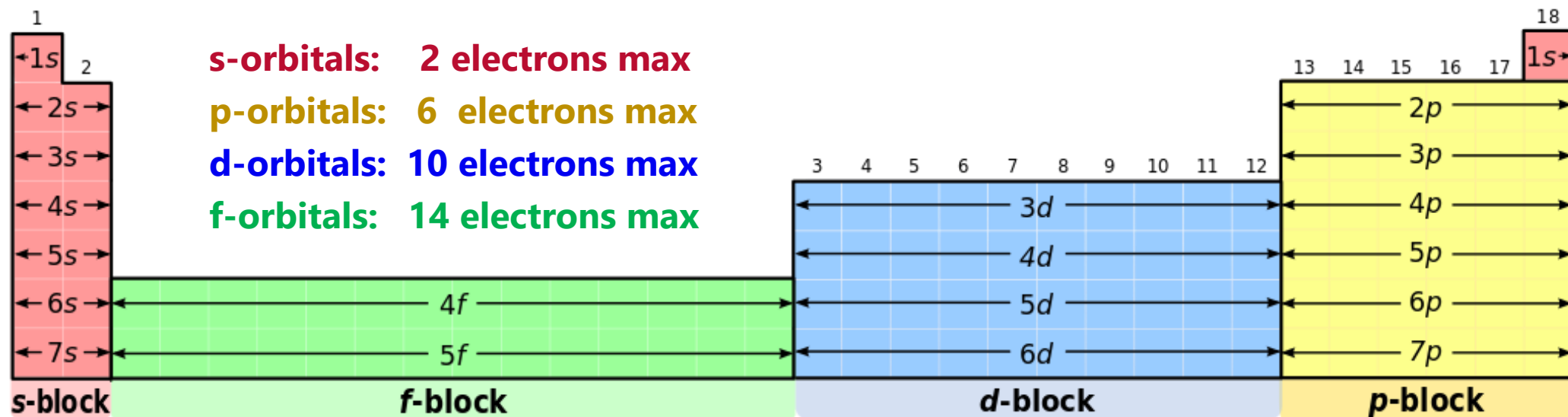
Atomic Weight
Chemical Symbol
Name



Atomic Orbitals

Electrons bound to atoms occupy **orbitals** that reflect their energy and spatial distribution.

- Electrons fill lower energy orbitals first before filling the next orbital (Aufbau Principle):
1s, 2s, 2p, 3s, 3p, 4s, 3d, 4p...
- Orbitals** are grouped into electronic shells, where elements lower on periodic table have more shells.



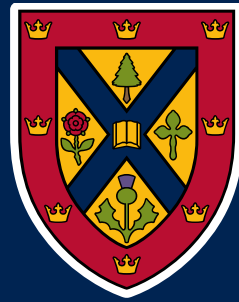
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Knowledge Development Questions

Elemental Atomic Properties

Consider hydrogen, lithium, carbon, nitrogen, oxygen, and chlorine when answering the following:

1. **Report** the atomic number of each element.
2. **Report** the group number of each element.
3. **Write** the electron configuration for each element.
4. **Look up** the electronegativity of each element and **order** them from most to least electronegative. **Comment** on whether the value you found agree with periodic trends for electronegativity.



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